

Professional Summary

Software engineer with 5+ years of experience spanning backend systems, database workflows, production reliability, and technical product delivery. I have led product and engineering decisions while designing and operating customer-facing order, payment, and real-time messaging systems where business requirements, API contracts, data consistency, observability, and deployment safety had to align. My strongest fit for Tesla is combining hands-on engineering with end-to-end delivery, cross-functional coordination, operational problem solving, and automation that supports business growth.

Fit for Tesla Sales & Delivery Applications

End-to-End Product Delivery Leading product direction, core algorithm improvement, and infrastructure/CI/CD readiness for a new consumer application launch.	Sales / Order Workflow Experience Built order creation, payment verification, refund policy, transaction handling, and status tracking flows.
SQL / Database-backed Systems Worked with PostgreSQL, Supabase, Oracle redo logs, data synchronization, and data pipelines.	Observability and Automation Implemented CloudWatch observability, blue-green deployment, internal diagnostic tooling, and automated development, testing, and documentation workflows.
Real-time System Design Deployed and operated an in-house chat service in production with load-tested scalability, CloudWatch observability, blue-green deployment, and no server outage over 2-3 months.	

Skills

Product / Program Delivery	Technical product ownership, requirements definition, prioritization, cross-functional coordination, release planning, end-to-end delivery
Backend / Data	NestJS, FastAPI, PostgreSQL, Supabase, REST API design, data modeling, transaction and payment state design, real-time messaging architecture, SQL querying and analysis
Frontend / Mobile	Flutter, React, Next.js, TypeScript, Riverpod, BLoC, React Query, responsive web, shared UI systems
Platform / Infra	AWS, AWS CDK, ECS Fargate, CloudWatch, GitHub Actions, Fastlane, blue-green deployment, Socket.IO, Redis Pub/Sub, Kafka, CI/CD, production observability
Automation	Python, C, Claude API, MCP, AI Agent Harness, Swagger-based code generation, test automation, documentation automation
Languages	Korean: Native English: Working proficiency (computer science coursework and technical documentation)

Professional Experience

Munto

2024.12 - Present

Full-stack Engineer

Owns backend and production-critical systems while contributing to product direction, cross-functional execution, and launch readiness across multiple consumer applications.

Web MAU 80K to 120K, total MAU 250K+

Production Reliability / Payment Flow

- Analyzed double-payment and payment-completed/order-missing risks caused by client-side payment confirmation and unstable order state transitions.
- Redesigned the flow so the server owns order creation, payment verification, transaction handling, refund policy, and final payment state decisions.
- Introduced a WAITING_PAYMENT state and order-level idempotency so retry and re-entry flows resolve against the same order instead of creating inconsistent payment attempts.
- Handled payment provider confirm timeouts with PROCESSING responses and order-status polling instead of prematurely failing user-visible payment flows.
- After rollout, no duplicate-payment incident attributable to the redesigned internal order/payment flow recurred; payment-provider incidents were isolated and handled separately.
- Supported payment operations for a service with approximately 15K total DAU and 8,000 weekly applications, currently processing KRW 600–700M in monthly transaction volume and reaching a peak of KRW 1B immediately after rollout.

Product Delivery / Application Architecture

- Acts as technical product owner for a new dating application launch, aligning product direction with marketing execution, improving the core matching algorithm, and overseeing infrastructure and CI/CD readiness.
- Separated server state with React Query cache and local UI state with hooks and Context API to reduce coupling between business data and screen state.
- Built and deployed an in-house chat service to production, replacing Sendbird, with separate REST/WebSocket deployments, PostgreSQL persistence, Redis Pub/Sub multi-node synchronization, message ordering, read-cursor unread counts, and push SLA design.
- Load-tested the architecture for workloads targeting up to 100,000 users and configured AWS CloudWatch observability and blue-green deployments; operated it in production without a server outage since June 2026.
- Migrated the first segment of approximately 500 active chat rooms, reducing SaaS cost by approximately \$300/month.
- Designed the munto-core common package for shared network, authentication, token, storage, logging, internationalization, Firebase, and chat SDK initialization modules.

Automation / Process Improvement

- Established a standard development process from planning/SRS to WBS, DBML, Swagger, TCL, implementation, and verification.
- Built an AI agent harness that reproduces the same development workflow across Claude, Cursor, and Codex, improving deliverable completion time by approximately 60%.

Cenacle Soft

2022.09 - 2024.03

Flutter Engineer

Built healthcare data and EMR-integrated application workflows from initial launch.

- Developed healthcare data, EMR integration, hospital reservation, and mobile patient chart workflows.
- Separated server DTOs from application models and designed factory-based domain models for heterogeneous health data.

WeDataLab

2020.04 - 2022.04

Software Engineer

Worked on DB monitoring solutions, system-level communication modules, internal tools, and Flutter applications.

- Developed an Oracle redo log-based data synchronization module for database monitoring and replication-related workflows.
- Managed Oracle Pro*C build and executable generation flows.
- Built socket communication and semaphore-based log transfer modules in C.
- Developed a Python/PyQt log viewer tool for internal inspection and debugging workflows.

Education

Sungkyunkwan University

2016.03 - 2023.08

B.A. in Philosophy, B.S. in Computer Science and Engineering

Keywords: technical product ownership, end-to-end delivery, enterprise applications, order/payment systems, SQL, database analysis, REST APIs, microservices, Redis, Kafka, Socket.IO, AWS, CI/CD, observability, system integration, automation, cross-functional programs.